Seminar: Signatures and Data Science

Applications of Signatures in Finance

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Existing Work

- **Data Generation** (Buehler et al. 2020 [2]): Using log-signatures and a Variational Autoencoder to generate paths of assets prices
- **Anomaly Detection & Classification** (Gyurkó et al. 2014 [5]): Classification of 30-minute time intervals from which Crude Oil Futures are sampled using Ridge Regression

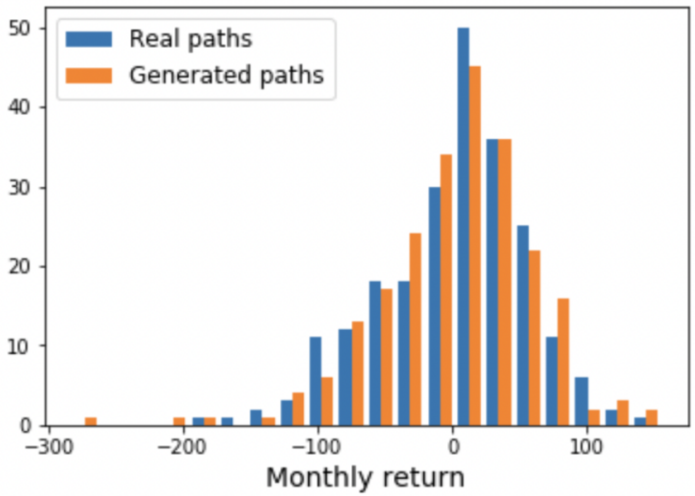


Figure: Generated returns vs actual returns of S&P 500[2]

Time	Accuracy
9:30-10:00	0.946
10:30-11:00	0.946
10:30-11:00	0.973
12:30-13:00	0.950
13:00-13:30	0.942

Table: Classification against 14:00-14:30 Time-frame[2]

Limit Order Books

- **Limit Order Books** (LOBs) are records of aggregated outstanding **limit orders** (orders submitted at a pre-determined price) and employed by electronic exchanges like the NASDAQ

Message book

Time(sec)	Price($)	Volume	Event Type	Direction
34203.011926972	585.68	18	execution	ask
34203.011926973	585.69	16	execution	ask
...	...	...	...	...
34203.011988208	585.74	18	cancellation	ask
34203.011990228	585.75	4	cancellation	ask
...	...	...	...	...
34203.012050158	585.70	66	execution	bid
34203.012287906	585.45	18	submission	bid
34203.089491920	586.68	18	submission	ask

Figure: Example of Message Book

	Size	Price
Asks	200	$80.03
	0	$80.02
	400	$80.01
	1100	$80.00
Bids	1000	$79.99
	500	$79.98
	50	$79.97
	400	$79.96

Figure: Example of Order Book[8]

Notation

For time horizon $T > 0$, $k \in \mathbb{N}$ and asset S

- **Bid-price process** $\left(P_t^{\text{Bid},k}\right)_{t\in[0,T]}$ is the k best (highest) bid price with corresponding **Bid-volume process** $\left(V_t^{\text{Bid},k}\right)_{t\in[0,T]}$ for asset S
- **Ask-price process** $\left(P_t^{\text{Ask},k}\right)_{t\in[0,T]}$ is the k best (lowest) ask price for S with corresponding **Ask-volume process** $\left(V_t^{\text{Ask},k}\right)_{t\in[0,T]}$ for asset S
- **Mid-price process** $(P_t)_{t\in[0,T]}$ with $P_t := \frac{P_t^{\text{Bid},1} + P_t^{\text{Ask},1}}{2}$
- **Order Book process** $(O_t^k)_{t\in[0,T]}$ of level k with

$$O_t^k := (P_t^{\text{Bid},1:k}, V_t^{\text{Bid},1:k}, P_t^{\text{Ask},1:k}, V_t^{\text{Ask},1:k}) \in \mathbb{R}^{4k}$$

Limit Order Book Models

- Modelling order books can help understand price formation at granular level (Cont et al. 2011 [3] suggests linear relationship between order flow imbalance and price changes) and can be used for data generation
- Different modelling approaches: Queue-based models, Agent-based models
- Order-driven markets (Jenssen 2011 [6]) generalizes conventional markets in financial mathematical, by including process for bid and ask

Problem: Universality Hypothesis (Sirignano and Cont [8])

Question: Does there exist a universal price mechanism? In other words, do there exist functions $F_t: \left(\mathbb{R}^{2k}\right)^{[0,t]} \times \mathbb{R} \to \mathbb{R}$ such that $P_{t+\Delta t} - P_t = F_t((O_s^k)_{s\in[0,t]}, \varepsilon_t)$ and do not depend on the asset S.

- Approach using Deep Learning: Sirigano and Cont [8] use Recurrent Neural Network $f(\cdot|\theta)$ parameterized by θ and suggest that

$$P_{t+\Delta t} - P_t = f((O_s^k)_{s\in[0,t]}, h_t|\theta)$$

- Approach using Signatures: Denote $S^m(X)$ as the truncated signature up to level m of path $X: [0,T] \to \mathbb{R}^d$ and let $s_d(m)$ be the dimension of $S^m(X)$. We propose the linear model

$$P_{t+\Delta t} - P_t = \left\langle \beta_m, S^m((O_s^k)_{s\in[0,t]}) \right\rangle$$

for parameter $\beta_m \in \mathbb{R}^{s_d(m)}$ and euclidean norm $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^{s_d(m)}$

Why use Signatures?

Theorem (Universal Approximation Theorem [4])

Let $D \subset BV\left(\mathbb{R}^d\right)$ be a compact set of paths with bounded variation and $X_0 = 0$ for all $X \in D$. Denote with $\tilde{X} = (X_t, t)_{t\in[0,T]}^{\top}$ the associated time-augmented path. Let $f : D \to \mathbb{R}$ be a continuous function. Then, for every $\varepsilon > 0$, there exist $m^* \in \mathbb{N}, \beta^* \in \mathbb{R}^{s_d(m^*)}$, such that

$$\left| f(X) - \left\langle \beta^*, S^{m^*}(\tilde{X}) \right\rangle \right| \leq \varepsilon$$

holds for any $X \in D$.

Classification Task

In order to evaluate and train the models, Sirigano and Cont [8] introduce classification task for predicting direction of the next price move.

Discretize time $t = 0, \dots, n$ and consider stream of order-book $(O_t^k)_{t=1,\dots,n}$ of level k

$$f((O_t^k)_{t=0,\dots,n}) = \begin{cases} -1 & \text{if } \frac{P_{n+1}-P_n}{P_n} \in (-\infty, -\alpha), \\ 0 & \text{if } \frac{P_{n+1}-P_n}{P_n} \in [-\alpha, \alpha], \\ 1 & \text{if } \frac{P_{n+1}-P_n}{P_n} \in (\alpha, \infty) \end{cases}$$

for a threshold value $\alpha > 0$.

Experimental Setup 1

Dataset: Benchmark LOB Dataset (Ntkaris et al. 2017 [7]) consisting of 5 different stocks on the NASDAQ Nordic exchange over 10 day trading period (01/06/2010-14/06/2010) and $\approx$ 4 million data points.

- Data sampled from 10-level LOB after 10 consecutive new events and already normalized
- For computational reasons, choose order book up to level 2, i.e. $(P_t^{\text{Bid},1}, P_t^{\text{Bid},2}, V_t^{\text{Bid},1}, V_t^{\text{Bid},2}, P_t^{\text{Ask},1}, P_t^{\text{Ask},2}, V_t^{\text{Ask},1}, V_t^{\text{Ask},2}) \in \mathbb{R}^8$

Setup: Choose $\alpha = 0.02$ and use 100 previous data-points to predict direction of next mid-price. Dataset $\mathcal{D} = ((X_1, Y_1), \ldots, (X_N, Y_N))$ with $X_i \in \mathbb{R}^{100\times 8}$ and $Y_i \in \{-1, 0, 1\}$

Data

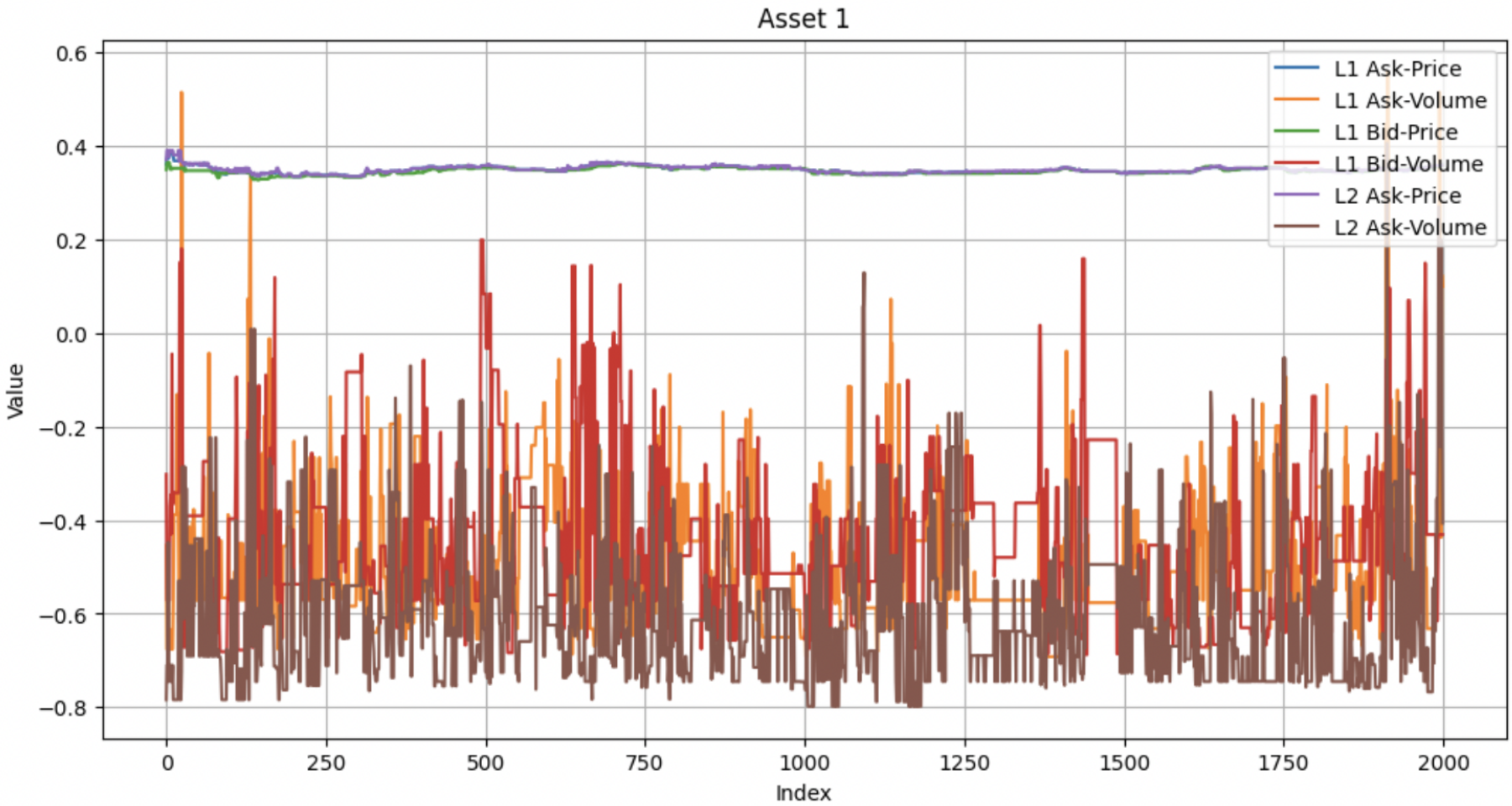


Figure: Time Series of Asset 1

Experimental Setup 2

Method: Use first 8 days as training data and last 2 days as test data.

- Single-Stock Model: Train model only on Asset 1. Training dataset 26439 samples with {-1: 27%, 0: 46%, 1: 27%}
- Universal-Stock Model: Train model on Assets $1, 2, 3, 4$. Training dataset 70000 samples with {-1: 17%, 0: 65%, 1: 18%}

Evaluation:

- Use F1-Score as metric
- Test data (20000 samples) from all 5 Assets.
- Only use Asset 5 (14747 samples)

Ridge Regression

Given dataset $\mathcal{D} = ((X_1, Y_1), \ldots, (X_N, Y_N))$ with $X_i \in \mathbb{R}^d$ and $Y_i \in \mathbb{R}$. Set $\mathbf{X} = (X_1, \ldots, X_N) \in \mathbb{R}^{N \times d}$ and $\mathbf{y} = (Y_1, \ldots, Y_N) \in \mathbb{R}^N$, then ridge regression minimizes the following loss function w.r.t. parameter $\beta \in \mathbb{R}^d$

$$\hat{L}(\beta) = \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda \|\beta\|^2$$

where $\lambda \geq 0$ is a regularization parameter. The solution is given by:

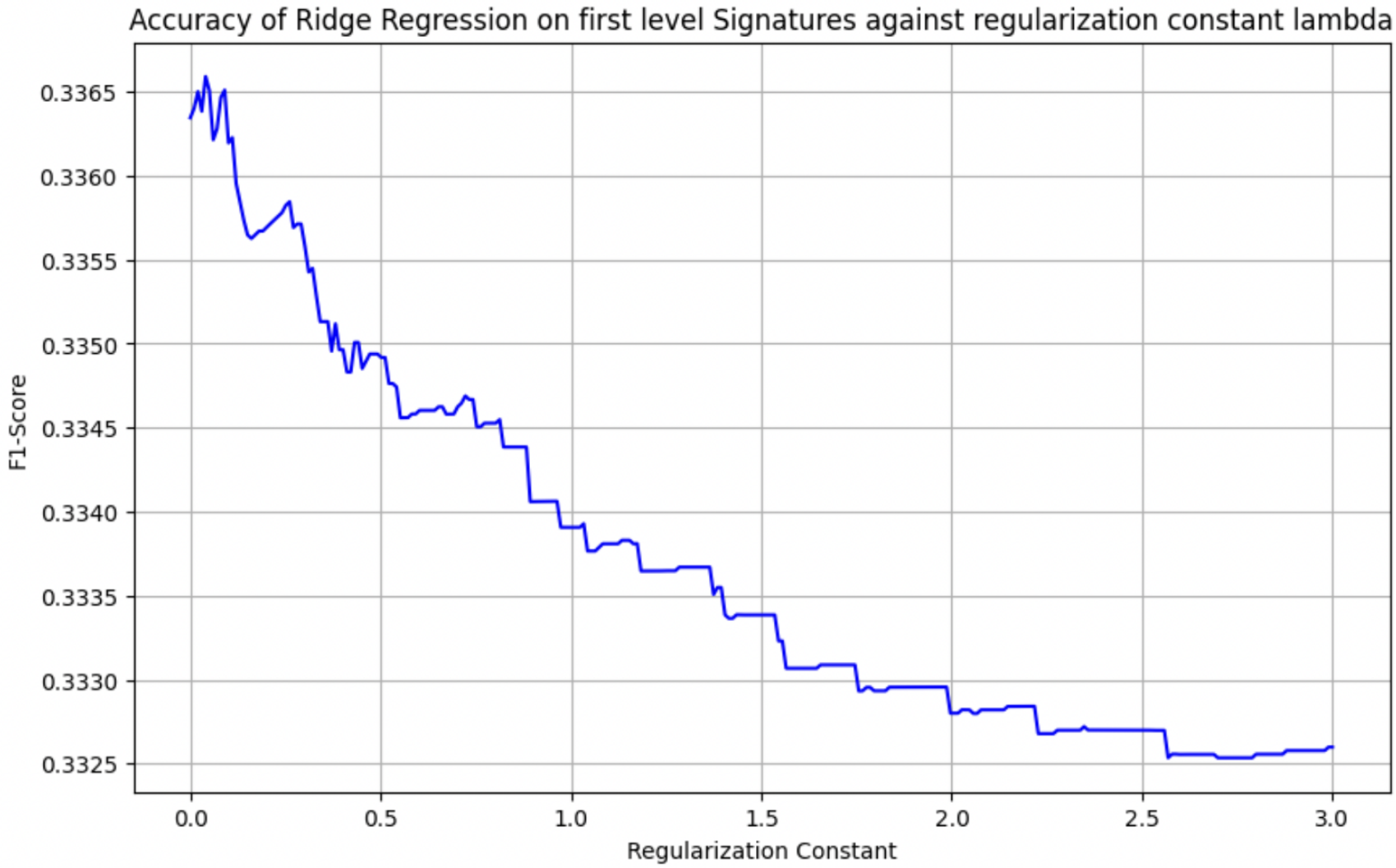
$$\beta = \left(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}\right)^{-1} \mathbf{X}^\top \mathbf{y}$$

Linear Signature model (Fermanian 2022 [4])

Require: Dataset $\{(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n)\}$, $\mathbf{X}_i \in \mathbb{R}^{M \times d}$, $Y_i \in \mathbb{R}$
Ensure: Signature degree $\hat{m} \in \mathbb{N}$ and parameter $\hat{\beta}_{\hat{m}} \in \mathbb{R}^{s_d(\hat{m})}$

1: Add time dimension to $\mathbf{X}_i = (X_1, \dots, X_d)$, i.e. $X^t_{d+1} := t$ for $t = 1, \dots, M$ and consider $\tilde{\mathbf{X}}_i = (\mathbf{X}_i, X_{d+1})$
2: Select the regularization parameter λ by cross validation on the regression model with $\{S^1(\tilde{\mathbf{X}}_1), \dots, S^1(\tilde{\mathbf{X}}_n)\}$ as predictors.
3: **for** $m = 1, \dots, \hat{M}$ **do**
4: Compute m-th truncated signatures: $\{S^m(\tilde{\mathbf{X}}_1), \dots, S^m(\tilde{\mathbf{X}}_n)\}$.
5: Fit Ridge regression and compute loss $\hat{L}_n(m)$.
6: Compute the penalization $\text{pen}_n(m) = K_{\text{pen}} \frac{\sqrt{s_d(m)}}{n^\rho}$.
7: **end for**
8: Choose $\hat{m} = \arg\min_{0 \le m \le \hat{M}} (\hat{L}_n(m) + \text{pen}_n(m))$.
9: Compute $\hat{\beta}_{\hat{m}}$ using Ridge Regression on $\hat{m}$-truncated signatures

Choosing Regularization parameter λ



Most significant signature

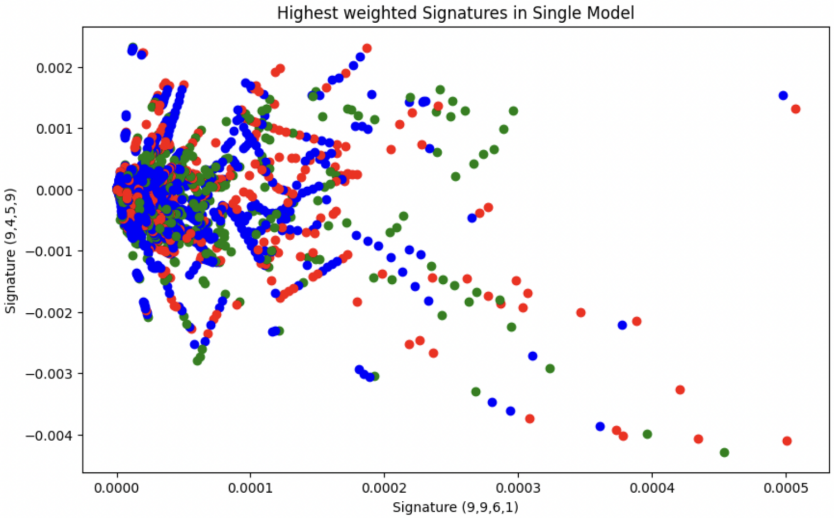


Figure: (9,9,6,1) against (9,4,5,9)

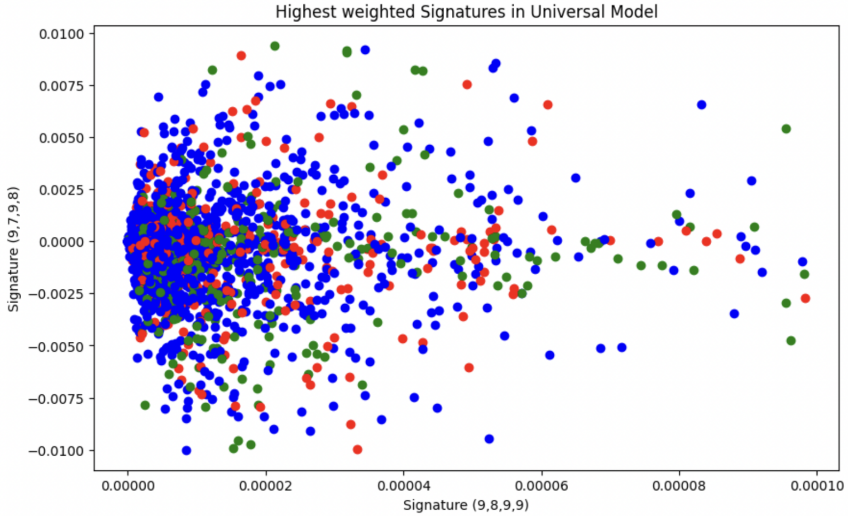


Figure: (9,8,9,9) against (9,7,9,8)

Log-signature with Neural Networks

Definition (log-signatures)

One can also define the signature of a path $X : [0, T] \to \mathbb{R}^d$ as a formal power series with indeterminates $e_1, \dots, e_d$

$$S(X) = \sum_{k=0}^{\infty} \sum_{i_1,\dots,i_k \in \{1,\dots d\}} S(X)^{i_1,\dots,i_k} e_{i_1} \dots e_{i_k}.$$

Then, the **log-signature** is defined as

$$\log S(X) := \sum_{n=1}^{\infty} \frac{(-1)^n}{n} (1 - S(X))^{\otimes n}$$

Log-signatures Dimensions

	$d=1$		$d=2$		$d=3$		$d=4$	
	$\sigma_{d,N}$	$\eta_{d,N}$	$\sigma_{d,N}$	$\eta_{d,N}$	$\sigma_{d,N}$	$\eta_{d,N}$	$\sigma_{d,N}$	$\eta_{d,N}$
$N=1$	1	1	2	2	3	3	4	4
$N=2$	2	1	6	3	12	6	20	10
$N=3$	3	1	14	5	39	14	84	30
$N=4$	4	1	30	8	120	32	340	90
$N=5$	5	1	62	14	363	80	1364	294
$N=6$	6	1	126	23	1092	196	5460	964

Figure: Dimensionality of Signatures vs. Log-Signatures [1]

Model Performance Comparison

Models	F1 Score on Test Set	F1 Score on 5th Asset
Benchmark 1	0.28	0.29
Benchmark 2	0.38	0.20
Single-Asset Signature Model (4)	0.37	0.43
Uni-Asset Signature Model (4)	0.33	0.30
Single-Asset NN with log-signature (3)	0.38	0.41
Uni-Asset NN with log-signature (3)	0.36	0.35
DeepLOB (Zhang 2020)	0.83	/

Table: F1 Scores of Different Models

Evaluation

- In [8], models accuracy were high and able to generalize well to unseen stocks, which suggests **Generalization**
- Prediction accuracy is not very high, but signature models are robust on assets not seen before in comparison to Benchmark 2
- In [8], Universal model outperforms single-asset model consistently by 2%, suggesting features are not asset-specific and suggests **Universality**
- Universal model performs significantly worse than single-asset model, may be due to normalization. However, [8] suggests data normalization does not improve accuracy.

[1] Christian Bayer et al. *Optimal stopping with signatures*. 2021. arXiv: 2105.00778 [math.PR]. URL: https://arxiv.org/abs/2105.00778.

[2] Hans Buehler et al. "Generating Financial Markets With Signatures". In: *SSRN* (2020). URL: https://ssrn.com/abstract=3657366.

[3] R. Cont, A. Kukanov, and S. Stoikov. "The Price Impact of Order Book Events". In: *Journal of Financial Econometrics* 12.1 (June 2013), pp. 47–88. ISSN: 1479-8417. DOI: 10.1093/jjfinec/nbt003. URL: http://dx.doi.org/10.1093/jjfinec/nbt003.

[4] Adeline Fermanian. *Functional linear regression with truncated signatures*. 2022. arXiv: 2006.08442.

[5] Lajos Gergely Gyurkó et al. "Extracting information from the signature of a financial data stream". In: *arXiv* (2014). URL: https://arxiv.org/pdf/1307.7244.

[6] Arend Jenssen. "Order book models, signatures and numerical approximations of rough differential equations". In: (2011). URL: https://www.ora.ox.ac.uk/objects/uuid:264e96b3-f449-401b-8768-337acab59cab.

[7] Adamantios Ntakaris et al. "Benchmark dataset for mid-price forecasting of limit order book data with machine learning methods". In: *Journal of Forecasting* 37.8 (2018). ISSN: 1099-131X. DOI: 10.1002/for.2543. URL: http://dx.doi.org/10.1002/for.2543.

[8] Justin Sirignano and Rama Cont. *Universal features of price formation in financial markets: perspectives from Deep Learning*. 2018. arXiv: 1803.06917 [q-fin.ST]. URL: https://arxiv.org/abs/1803.06917.